Homework 3 Individual

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1 Question 1.

Weight only needs the single subscript i because that is going to remain constant through each sweat measurement, so they only need to account for it on the subject level. The no-tattoo indicator, on the other hand, will change despite the subject being the same, so this is on the observation level.

2 Question 2.

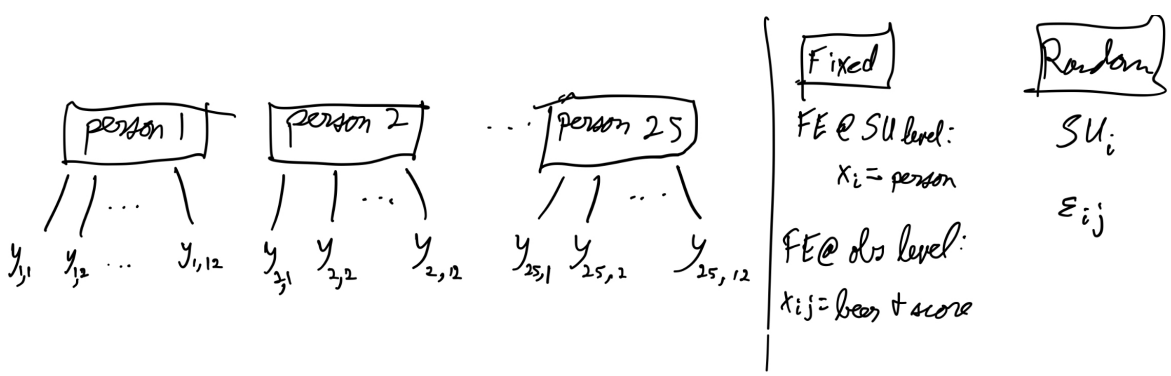
If the classmate used age_{ij} despite it being measured one per subject, at the observation level, *modeldiagramR* uses data to generate a corrected model by creating a model diagram based on the data, which would make this student's mistake clearly visible.

3 Question 3.

I'd use the LearnVizLMM package to check the model diagram fro lab 7 before cleaning and wrangling the data because it doesn't require the data or a model object to draw the diagram.

4 Question 4.1

Measurement structure:



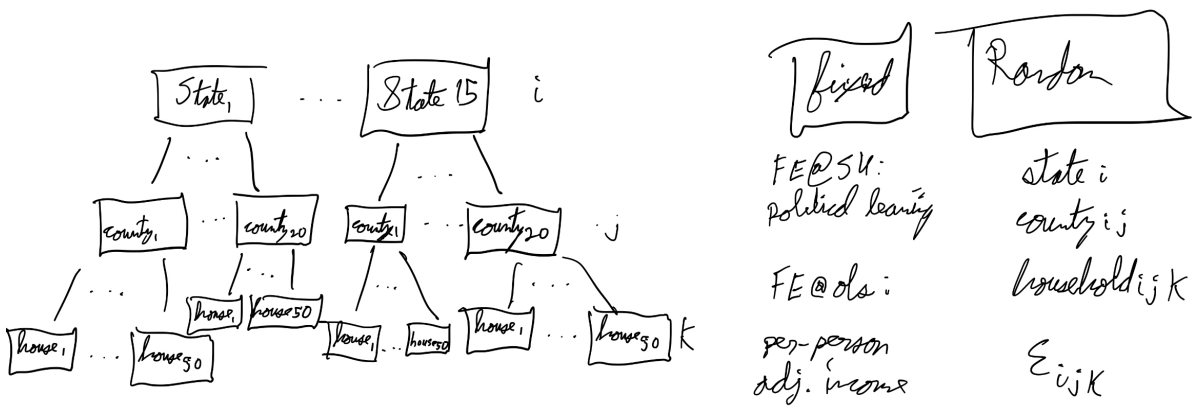
- fixed effects column:
  - people @SU level (x_i)
  - Time they took quiz, liters of beer consumed @obs level (x_{ij})

- random effects column: $\mathrm{SU_i}$, ϵ_{ij}
- i = 1-25 (index people)
- j = 1-12, (index observation within people (number of tests taken))

Theoretical model:
$\mathrm{IQScore}_{ij} = \mu_{ij} + Person_i + \epsilon_{ij}$
$\mu_{ij} = \beta_0 + \beta_1 I_{test2} + \ldots + \beta_{11} I_{test12} + \beta_{13}\mathrm{beer_liters}$
$SU_i \sim \mathcal{N}(0, \sigma^2_{person})$
$\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$
Model params:

- β_0: expected IQ for someone who doesn't drink
- β_i = the change in intercept for test i compared to baseline
- β_{13} = the change in intercept compared to baseline given the liter of beer they drank in the last two weeks.

5 Question 4.2



- i = 1-15 (index state)
- j = 1-20 (index county)
- k = 1-50 (index households)

Theoretical model: $\mathrm{diff}_{ijk} = \mu_{ijk} + state_i + county_{ij} + household_{ijk} + \epsilon_{ijk}$
$\mathrm{Diff}_{ij} = \mu_{ijk} + State_i + county_{ij} + household_{ijk} + \epsilon_{ijk}$
$\mu_{ijk} = \beta_0 + \beta_1 \mathrm{per\ person\ adj\ income} + \beta_2 I_{right_leaning} + \beta_3 \mathrm{youngest}$
$state_i \sim \mathcal{N}(0, \sigma^2_{state})$
$county_{ij} \sim \mathcal{N}(0, \sigma^2_{count})$
$household_{ijk} \sim \mathcal{N}(0, \sigma^2_{household})$
$\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$
Model params:

- β_0: mean adjusted household income for left-leaning if youngest per-person-adj income is zero and youngest age in household is zero
- β_1: how the intercept changes with the inclusion of per-person-adjusted-income compared to baseline

- β_2: how the intercept of baseline changes based on whether the county was right-leaning or not
- β_3 how the intercept of baseline changes based on age of the youngest person in the household